

SLM-125M — Completed Build Pipeline

From-scratch 125.8M-param Llama-style legal / financial SLM · 12L / 768d / 12h · 16,384 vocab · 1,024 context · 2026-09-07

1 · DATA PIPELINE

Phases 0-4 · CPU · ~\$0.18

Smoke test + per-source token-yield measurement

Stream & deterministic clean - 718K -> 698K docs kept

MinHash / LSH near-dedup + exact dedup + CaseHOLD / LexGLUE decontamination -> ~670K docs

Train fresh 16,384 byte-level BPE tokenizer on the corpus

Tokenize + pack into 1024-token uint16 windows, 99 / 1 train-val split

RESULT

2.04B train tokens
1,991,282 train windows
realized mix: SEC 42% / case-law 35% /
FineWeb-Edu 23%

2 · PRETRAINING

Phase 5 · 8xH100 DDP (Modal) · ~\$8-9

1 epoch · 3,877 steps · ~2.03B tokens seen · 524K-token global batch

AdamW (beta 0.9 / 0.95, wd 0.1), LR 6e-4 -> 6e-5 cosine, 200M-token warmup

Completed in 2 runs - hit \$10 budget cap at step 3064, resumed from checkpoint

RESULT

train loss 9.107 -> 2.4832

3 · BASE MODEL DEPLOY

Phase 6

Push base model + tokenizer to the Hugging Face Hub

RESULT

HF: IndraniBera/slm-125m-base (private)

4 · QA SUPERVISED FINE-TUNING

Phases 7-10 · 1xL4

Generate 1,196 QA pairs with gpt-5.4-nano, grounded in pretrain-corpus excerpts

Domain split: Legal 656 / SEC 300 / General 240

Tokenize into loss-masked, padded SFT examples

SFT-train from the base checkpoint · 2 epochs · LR 5e-5

Base-vs-SFT head-to-head on CaseHOLD (held out of pretraining)

RESULT

assistant-token perplexity 8.11
CaseHOLD accNorm 0.170 -> 0.166
HF: IndraniBera/slm-125m-sft (private)

5 · RAFT (retrieval-aware fine-tuning)

Phases 11-12 · 1xL4 · ~\$0.14 · continues from the SFT checkpoint

Reformat the same 1,196 pairs with 3 retrieved-style passages each

1 golden + 2 distractors ~79% of the time · 3 distractors, no golden ~21%

golden-present 942 / golden-absent 254

Tokenize RAFT examples (mirrors the SFT tokenizer exactly)

RAFT-train from the SFT checkpoint · 4 epochs · 296 steps · LR 3e-5

RESULT

assistant-token perplexity 2.821
HF: IndraniBera/slm-125m-raft (private)

6 · RLAIF — DPO preference tuning

Phases 14-17 · 1xL4 · ~\$4 · branches from the SFT checkpoint (parallel to RAFT)

Sample 3 on-policy SFT answers per training prompt (temp 0.8) as preference candidates

gpt-5.4-mini judge, 8-shard parallel, order-swapped + confidence-filtered (≥ 0.6)

On-policy Track A dropped - 125M samples too weak to rank; 1,076 Track-B pairs kept

Track B: grounded gold answer vs. targeted corruption (truncate / unsupported / wrong entity)

DPO-train from the SFT checkpoint · 4 epochs · 536 steps · LR 1.5e-5 · beta 0.05 · +0.1 aux-NLL

Held-out win-rate eval - DPO vs SFT on 120 prompts, gpt-5.4-mini judge

RESULT

reward margin +0.03 -> +1.3 (healthy DPO curve)
win-rate vs SFT 0.525 (target ≥ 0.60 - not met)
assistant-token perplexity 8.11 -> 8.97
HF: IndraniBera/slm-125m-dpo (private)
verdict: no measurable gain - 125M capacity limit

7 · RAG INDEX

Phase 13 · CPU

Build BM25 lexical inverted index (in-memory, no vector DB)

6,000 chunks - legal 3,300 / SEC 1,500 / general 1,200

~880-char paragraph-snapped chunker (same noise profile as RAFT training)

RESULT

42,127 unique terms
retrieve k = 3

8 · FRONTEND & SERVING

Vercel + Modal FastAPI endpoints

Next.js app deployed on Vercel (project slm-125m)

/ base-model completion playground + architecture / corpus / loss-curve viz - LIVE

/sft QA playground (SFT model) + dataset & eval stats - built locally, deploy pending

/rag RAG playground - RAFT model + live BM25 retrieval - built locally, deploy pending

Modal inference endpoints: Playground (live) · PlaygroundSFT · PlaygroundRAG

RESULT

LIVE: <https://slm-125m-psi.vercel.app>
base playground deployed; /sft + /rag not yet pushed

NOTE

RLAIF / DPO (stage 6) was run end-to-end and the checkpoint published, but did not clear the 0.60 win-rate bar - it is not a strict upgrade over SFT.